Sravan Potnuru

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Professional Summary

AI/ML Engineer with nearly 2 years of experience building production machine learning, deep learning, computer vision, and Generative AI systems. Shipped a real-time video analytics platform processing 20+ live CCTV streams (YOLO, InsightFace, NVIDIA Triton, AWS Kinesis Video Streams) and cut end-to-end latency 83%. Builds RAG and knowledge-graph pipelines (Neo4j, FastAPI) for a US client's enterprise AI, and explores LLM post-training (LoRA, CPT) and agentic AI in personal R&D. Also builds full-stack applications (React, Spring Boot) in personal projects and published a pip-installable LLM observability SDK on PyPI.

Education

Rajiv Gandhi University of Knowledge Technologies (RGUKT) IIIT Nuzvid, Nuzvid, Andhra Pradesh *2021 – 2025*
Bachelor of Technology (B.Tech) in Computer Science Engineering GPA: 9.0/10

RGUKT IIIT Nuzvid *2019 – 2021*
Pre-University Course (PUC) – Intermediate GPA: 9.62/10

Skills

Languages & Frameworks: Python, Java, SQL, PyTorch, TensorFlow, Keras, OpenCV, pandas, YOLO, InsightFace, BoT-SORT, Machine Learning, Deep Learning, Computer Vision, Object Detection, Generative AI, Natural Language Processing (NLP), Large Language Models (LLMs), LLM Fine-Tuning, LoRA, Continued Pre-Training (CPT), Retrieval-Augmented Generation (RAG), AI Agents (LangGraph), Model Context Protocol (MCP), FastAPI, Spring Boot, React JS, REST APIs, HTML, CSS, Bootstrap
Cloud Platforms: AWS (EC2, S3, Lambda, Kinesis Video Streams), Azure (Data Factory, DevOps), Akash Network (GPU)
DevOps & Tools: MLOps (model serving and deployment), NVIDIA Triton Inference Server, TensorRT, Inference Optimization, Docker, GitHub Actions (CI/CD), Ruff, Apache NiFi, Databricks, Locust (load testing), Git, Postman, Microsoft Power Automate, Microsoft Teams, Microsoft Copilot
Databases: Vector Databases (Qdrant, pgvector), MySQL, PostgreSQL, Neo4j (Cypher)

Professional Experience

Parabola9, AI/ML Engineer *Dec 2024 – Present*

- Co-developed a real-time video analytics platform processing 20+ live CCTV streams at 5 FPS with 10-second end-to-end event latency, using FastAPI microservices and GPU-based computer vision inference.
- Fully automated footage ingestion for 20+ cameras (zero manual steps) with a serverless pipeline on AWS Lambda, Kinesis Video Streams, and S3.
- Cut end-to-end latency 83% (60 s to 10 s) by serving YOLO and InsightFace models on NVIDIA Triton Inference Server with TensorRT, FP16, and dynamic batching across 20+ concurrent streams.
- Built a PyTorch computer vision pipeline (YOLO, BoT-SORT, InsightFace, OpenCV) for real-time object detection, person tracking, and face recognition, reaching 80% accuracy.
- Implemented face recognition matching and clustering with face embeddings stored in a vector database.
- Containerized and deployed 5+ AI microservices with Docker on AWS EC2.
- Delivered a real-time warehouse surveillance system detecting PPE violations and security events across 20+ cameras.
- Collaborate with the CTO and co-founders on project planning, architecture decisions, and status updates.
- Reduced infrastructure cost by serving production GPU inference on Akash Network (Linux/AMD64).

Parabola9 – US Client Project: Rich's, AI/ML Engineer *Feb 2026 – Present*

- Built Azure Data Factory ETL pipelines ingesting Azure DevOps work items and Azure File Share documents into Bronze/Silver/Gold layers, expanding AI knowledge coverage.
- Built FastAPI APIs for enterprise document ingestion and semantic retrieval, grounding Generative AI answers for internal business users.
- Building a RAG assistant over SAP data and enterprise documents for the client's Enterprise AI Harness, built on a semantic layer and knowledge graphs.
- Redesigned Neo4j knowledge-graph pipelines (Cypher) powering enterprise RAG search across internal documentation, improving retrieval relevance.
- Designed an in-app survey and feedback feature modeled on Claude's session feedback prompt ("How is Claude doing this session?").

- Developed persona routers, based on the plugin concept, that automatically route users to the right persona from their profile.
- Contributed to integrating the applications with Microsoft Teams.
- Converted the product into Model Context Protocol (MCP) servers and exposed them to Microsoft Copilot.
- Evaluated LLM-backed APIs with Locust load tests, measuring p50, p95, and p99 latency and error rates to find performance bottlenecks before production release.

Personal Projects

LogIQ – LLM Observability Platform

- Built a multi-tenant AI observability platform (FastAPI, React, PostgreSQL) with a pip-installable Python SDK published on PyPI and multi-LLM support (OpenAI, Anthropic, Azure, Ollama).
- Implemented log ingestion, root-cause analysis with automatic rule-based fallback, a debugging chat assistant that identifies the source error file, multi-channel alerting (Slack, Email, and Microsoft Teams via Microsoft Power Automate workflows), and cloud monitoring (AWS/Azure/GCP).
- Validated LogIQ end-to-end on a real project.

Personal R&D (In Progress)

- Ran LLM post-training on a Qwen 2B model using RunPod GPUs: supervised fine-tuning (SFT) did not meet the goal, so shifted to continued pre-training (CPT) on the base model, raising evaluation accuracy from 38% to 44%.
- Analyzed the causes of the limited accuracy gain and tuned the training recipes; retraining with revised configs is in progress.
- Building AI Operator, an AI agent that audits work against company rules and makes changes through an approval workflow.
- Researching an organization-level memory layer for enterprise AI, with access control modeled on SharePoint ACLs, and drafting its implementation plan.
- Set up GitHub Actions CI/CD pipelines for personal projects with Ruff linting, automated CI checks, security-issue flagging, and a CD deployment stage.

ParaBot – AI File and Content Summarizer

- Built an AI assistant (FastAPI, React, OCR, Gemini) that processes text, images, PDFs, and website content with summarization and Q&A.

ScreenSaga – OTT Platform

- Built a full-stack OTT platform (React JS, Spring Boot REST APIs, MySQL) with movies, web series, trailers, personalized watchlists, Continue Watching, season/episode data, and a feedback system.
- Implemented secure authentication (email verification, password reset, account deletion) with JSON APIs and SMTP-based email confirmations.

Honors & Leadership

- Achieved NCC “B” and “C” Certificates.
- Received Certificate of Participation and Certificate of Appreciation, National Level Techno Management Fest.